



**PX 55**

**E-1379**

## **Technical Documentation**



## Table of Contents

### **1. Safety Instructions**

#### **1.1 Safety warnings**

### **2. Compliance Declaration**

#### **2.1 EU declaration**

### **3. General Information**

#### **3.1. Quality specification of water - 553406 R10**

#### **3.2. Quality specification of air - 553407 R6**

### **4. Process Information & Drawing**

#### **4.1. P&ID - 9680029245 R01**

### **5. Installation Manuals**

#### **5.1. Instruction book - 9034330-02 R01**

### **6. Item List/Component List**

#### **6.1. Common Items - 9680033985 R00**

#### **6.2. Specific Items - 9680033999 R00**

#### **6.3. Hoses & Compensators - 9680029248 R01**

### **7. Mechanical Engineering Drawings**

#### **7.1. Installation drawings - PX 55**

#### **7.2. External connection list - PX 55**

#### **7.3. Drawing index - 9680029544 R03**

### **8. Component Documentations**

#### **8.1 Common Items**

- 8.1.1 Pressure Gauge - R41459 - Tempress
- 8.1.2 Non return valve - LKC-2 - Alfa Laval
- 8.1.3 Ball valve - Fig 177 - Weyn-Lauwer
- 8.1.4 Sight glass - 7005 - Kieselmann
- 8.1.5 Pressure Indicator - 18 - Norgren
- 8.1.6 Pressure reducing valve - R07 - Norgren
- 8.1.7 Piston valve - Uniflux 1014 - Evian
- 8.1.8 Y-Strainer - Fig 16L - Spirax Sarco
- 8.1.9 Flow restrictor - REG12 - Kobold
- 8.1.10 Non-return valve - 290X - Socla
- 8.1.11 Needle valve - Fig 226 - Econosto
- 8.1.12 Operating water tank - Alfa Laval

## **8.2. Specific Items**

- 8.2.1 Regulating valve - CPMI-2 - Alfa Laval
- 8.2.2 Sight glass light - BKVLR - Max Muller
- 8.2.3 Pressure transmitter - PTP35 / PTC31 - E&H
- 8.2.4 Sight glass - 7005 - Kieselmann
- 8.2.5 Flow switch - SWK2240 - Kobold

## **8.3. Hoses & Compensators**

- 8.3.1 Hoses - Interlink Marine
- 8.3.2 Hoses - Interlink Marine

## **9. Spare Parts Catalogue**

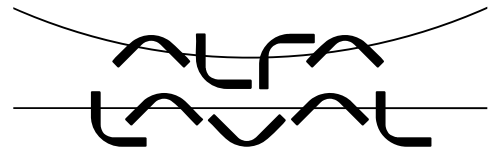
- 9.1. Spare part manual - 9034329-02 R01

## **10. Decommissioning**

## **11. Contact Alfa Laval Service**

# **1. Safety Instructions**

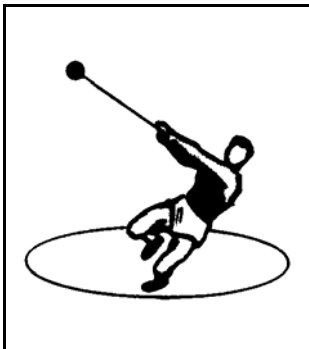
## **1.1 Safety warnings**



# SAFETY WARNINGS

## FOR HIGH SPEED SEPARATOR SYSTEMS

*All Personnel involved in the operation, maintenance or repair of separators or separation plants must read this manual thoroughly. This is essential for their safety.*

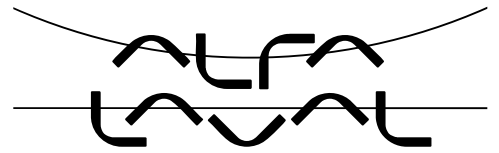


*The forces created due to the size, weight and high speed of the rotating components of a centrifugal separator are extremely great. Incorrect operation, maintenance and assembly, or use of non-original parts can cause serious damage to property and injury to personnel.*

### CONTENTS

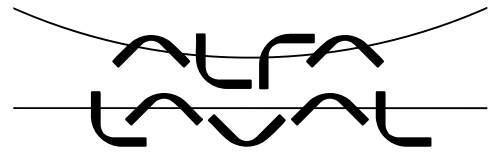
- |                                                                     |                                              |
|---------------------------------------------------------------------|----------------------------------------------|
| 1. General requirements                                             | 6. Stopping                                  |
| 2. Project planning and installation                                | 7. Dismantling, assembly and manual cleaning |
| 3. Pre-checks for the initial start of new or overhauled separators | 8. Maintenance                               |
| 4. Checks before and during each start                              | App.A Bowl types                             |
| 5. Operation                                                        |                                              |

**WARNING: NON-COMPLIANCE MAY CAUSE A SERIOUS ACCIDENT!**



# 1. GENERAL REQUIREMENTS

- 1.1 ALWAYS observe the Safety Warnings exactly.
- 1.2 ALWAYS follow the instructions in the Instruction Manuals carefully concerning installation, assembly, operation and regular maintenance.
- 1.3 NEVER operate a separator until it is properly mounted.
- 1.4 Ensure that all operators who operate and service a separator are well-trained and knowledgeable concerning the machine and the work to be carried out.
- 1.5 NEVER neglect alarms. Find out what caused the alarm and remedy it before the alarm is reset. 
- 1.6 ALWAYS use genuine Alfa Laval spare parts and tools.
- 1.7 ALWAYS follow local Safety Regulations concerning lifting devices, electric equipment, pressure vessels, inflammable, toxic or corrosive process liquids, etc.
- 1.8 Place information and warning notices in prominent places.
- 1.9 NEVER use the separator for separating liquid which is more corrosive or erosive or has higher density, higher temperature, different characteristics of the solids, etc., than originally specified. If your needs change, consult your ALFA LAVAL representative.
- 1.10 A separator has maximum and minimum operating temperature limits. Operation above or below these temperatures will reduce bowl strength. A weakened bowl may disrupt with grave risks to people and the surroundings. Therefore, strictly abide by the instructions concerning operating procedures that can alter bowl temperature (e.g. cooling time after sterilization or running a separator for long time in stand-by without cooling).
- 1.11 Separators for high temperature use and sterilizable separators have uninsulated surfaces which can cause burns if touched. 
- 1.12 NEVER work on mechanical equipment such as valves or pumps with the electric power or instrument air supply on. There is a mechanical hazard. 
- 1.13 NEVER work on electrical equipment with the main power supply on. There is an electric shock hazard. 
- 1.14 ALWAYS set the main switch to OFF and lock it in this position if you are going to disassemble the separator or parts of the auxiliary equipment. 



## 2. PROJECT PLANNING AND INSTALLATION

- 2.1 ALWAYS check with an Alfa Laval representative that the separator you plan to buy or use is suitable for the application. Elements that are important to consider are, e.g., process liquid density and temperature, corrosion characteristics, sludge characteristics, solids contents of the process liquid, toxicity, inflammability, degree of automation, etc.
- 2.2 ALWAYS check with an Alfa Laval representative that the installation is properly planned. Things that should be considered are, e. g.:
- A clear area around the equipment is required for servicing.
  - A lifting device of sufficient capacity and that can be positioned exactly over the centre of the bowl spindle is needed to remove the bowl assembly.
- 2.3 Ensure that all service media (compressed air, electric power, operating, cooling, and safety liquids, etc.) required for the separator have the correct quality and capacity. In cases where supply failure of the service media might cause damage, automatic supervision is recommended. This applies to operating media for solids discharging separators where a failure could cause unbalance and/or the cooling water for mechanical seals where a fault could cause hot spots or water leakage into the lubricating oil.
- 2.4 Ensure that the foundations are strong enough for the separator.
- 2.5 If the starting equipment containing the main switch is placed in such a position that it cannot be seen from the separator, most local safety regulations require that a separate safety switch must be installed in the power supply to the separator motor.
- 2.6 As the separator stands on soft rubber mountings, ALWAYS design the pipe-work to and from the separator in such a way that there will be sufficient flexibility. Possible vibration should not be allowed to be transmitted via the piping.
- 2.7 If the local safety regulations prescribe that the installation has to be inspected and approved by responsible authorities before the separation system is put into service, consult with such authorities before installing the equipment and have the projected installation approved by them.
- 2.8 NEVER transport or lift a separator with its bowl installed as bearings and the bowl spindle could be damaged.
- 2.9 Only motors (including necessary monitoring equipment) specified by Alfa Laval may be fitted to the separator. Violation of this can cause severe damage and create a safety risk for the personnel. Motor drives of variable speed must not be used unless approved



by Alfa Laval and used in combination with appropriate safety devices.

- 2.10 Design the system in such a way that there is always an open outlet for liquid from the bowl casing drain outlet.
- 2.11 If the sediment from the separator is discharged into a tank, this tank must be sufficiently ventilated. The connection between the separator and the tank must be of the size and configuration specified.

#### **For nozzle separators**

- 2.12 ALWAYS use strainers in the feed lines for the process liquid, cleaning liquid and water. Their mesh size must be smaller than the nozzle diameter.
- 2.13 Design the system in such a way that the feed can never be accidentally cut off while the bowl is rotating. A safety water/liquid supply of sufficient flow rate must be provided so that (to prevent unbalance) the bowl can be kept filled if the ordinary supply fails. **(This does not apply to UX type separators).**

#### **For separators with paring tube for concentrate discharge (e.g. UX type)**

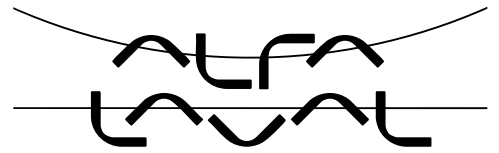
- 2.14 ALWAYS design the concentrate outlet line in such a way that there is an open outlet for the concentrate. If the separator operates against a closed outlet damage may occur.

#### **For separators equipped with a cooling jacket on the bowl casing or cyclone**

- 2.15 Ensure that the pressure of cooling liquid in the jackets cannot exceed the specified limit. Excess pressure will cause damage.

#### **For separators working in a closed system**

- 2.16 If the solids are discharged from the separator bowl casing into a closed system, ensure that this system cannot be overfilled or closed in such a way that the solids cannot leave the bowl casing. This could cause a hazardous situation.
- 2.17 Ensure that the gas pressure inside the bowl casing stays within specified limits. Otherwise the bowl casing might disrupt.



### 3. PRE-CHECKS FOR THE INITIAL START OF NEW OR OVERHAULED SEPARATORS

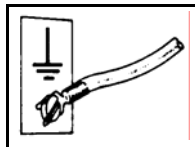
- 3.1 Check that installation of the rubber vibration dampers between the frame and foundation has been carried out according to the instructions.

- 3.2 Make sure that the drive unit contains the correct quantity of suitable oil.

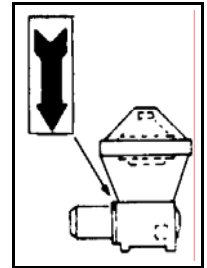


- 3.3 Check that all electric supervision signals and interlocks are connected and that they operate correctly.

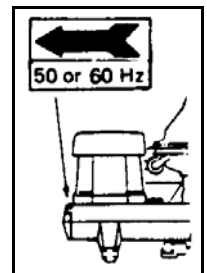
- 3.4 Make sure that the separator frame, control boxes and cabinets are connected to earth (ground) in accordance with local regulations. This is especially important in hazardous areas to reduce the risk of ignition due to static electricity.



- 3.5 Make sure that the motor rotates in the direction indicated by the arrow on the separator frame. If the bowl rotates in the wrong direction, the left-hand threaded lock ring(s) may unscrew. This could cause a serious accident!



- 3.6 Make sure that the frequency and voltage of the current to be connected is in accordance with separator specifications. Too high a bowl speed can create danger!

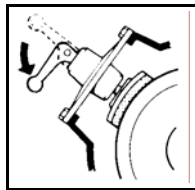


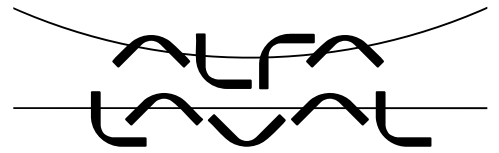
- 3.7 Check that the actual operating rpm with an empty bowl is the same as that specified in the Operator's Manual.



## 4. CHECKS BEFORE AND DURING EACH START

- 4.1 NEVER operate a separator if screws, nuts or other parts are missing, worn or damaged. Check that all clamps, pipe couplings and other fastenings have been securely tightened.
- 4.2 NEVER start the separator without being sure that the bowl is clean.
- 4.3 If the separator is equipped with a vibration alarm unit, check the setting and do not set it to a higher level than recommended. The pre-warning signal level may be adjusted to individual process conditions.
- 4.4 ALWAYS check that the brake is in the OFF position before a start (if applicable).
- 4.5 Check that all lines for service media are open and that safety liquid is available.
- 4.6 Supervise the entire starting cycle. If starting time is longer or power consumption higher than usual, stop the separator. The cause must be found and action taken to correct the fault before a restart.
- 4.7 To avoid overheating the motor, do not restart the separator more frequently than permitted in the applicable Operator's manual.



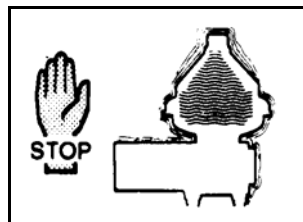


## 5. OPERATION

5.1 Wait until the bowl has reached its correct operating speed before the process liquid is turned on.

5.2 Check that there is no leakage from pipe connections on or around the separator.

5.3 If there is an unusual noise or vibration, stop the separator.



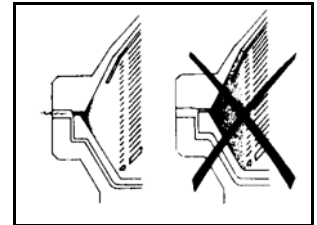
5.4 As the separator bowl rotates it is heated by air friction. Normally, this causes only a slight increase of the process liquid temperature, but, if the liquid flow is interrupted, the liquid remaining in the bowl will be heated to a temperature in excess of 100°C. This can cause a dangerous situation if the process liquid is inflammable or can decompose.  
In such cases, ensure that the flow to the separator is not interrupted long enough to cause a dangerous increase in temperature.

**For separators with solids discharge nozzles placed close to the periphery of the bowl (e.g. SX, QX, TX, DX types)**

5.5 ALWAYS run and stop the separator with a sufficient flow of liquid into the bowl to keep it filled. This reduces unbalance should nozzle blockage occur.

**For separators with intermittent solids discharge from the periphery of the bowl (e.g. PX, AX types)**

5.6 The solids are discharged from the bowl at intervals, the length



of which are dependent upon feed rate, solids content of the process liquid and characteristics of the solids. To avoid unbalance and risk of damage the solids must be discharged before they become packed hard or the solids space becomes overfull.

Before introducing the process liquid, check that the discharge interval time is set at a suitable rate and the discharge volume, if adjustable, is set at a suitable size. Operating times between discharges has a maximum limit determined by the compacting of the solids and a minimum limit for mechanical reasons or motor overload.

The discharge volume must be large enough to avoid incomplete emptying of solids from the bowl.

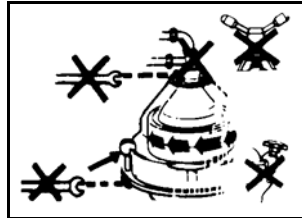
5.7 Do not adjust the discharge system in such a way that the solids will become so thick and dry that there is a risk of clogging the bowl cover and/or the cyclone outlet. Serious damage may occur if the solids cannot escape.

5.8 A separator with a variable discharge programme used for partial discharge should NEVER be changed to total discharge before consulting your ALFA LAVAL representative.

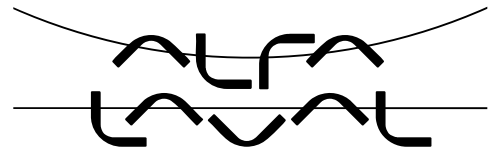


## 6. STOPPING

- 6.1 ALWAYS stop the separator with the bowl full of liquid.
- 6.2 If the separator is not equipped with an automatic emergency shut-down procedure, a manual procedure suitable for your installation must be used. Normally this means that if unusual vibration occurs the separator must be stopped with the feed (or safety water) on and the brake applied.
- 6.3 NEVER loosen any part of the separator and do not open inspection or cleaning hatches until the bowl has come to a complete standstill. Standstill is indicated by a lamp on the control panel and/or that the revolution counter is not rotating.



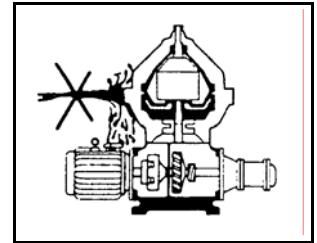
- 6.4 If the separator is equipped with an automatic safety liquid system do not shut off the main power to the system until the bowl has stopped and the manual valve for safety liquid has been closed.
- 6.5 NEVER feed any liquid to a stationary bowl.
- 6.6 Do not leave the bowl idle for any length of time with corrosive liquids in it.
- 6.7 NEVER let a separator stand still more than a few weeks (less in an area with high floor vibrations) without removing the bowl.
- 6.8 If the separator has been stopped because of vibration, do not use it again until the cause of vibration has been established and corrected.



## 7. DISMANTLING, MANUAL CLEANING AND ASSEMBLY

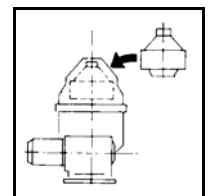
- 7.1 Check that the bowl is stationary, that the main power switch is turned off and, if possible, locked or fuses removed before starting any dismantling work. Hang up a warning sign to prevent the power being turned on.
- 7.2 ALWAYS use the special tools supplied by Alfa Laval when working on a separator. The use of improper or makeshift tools can cause accidents.
- 7.3 Position the hoist exactly when dismantling and assembling. Do not use a hoist that works jerkily or too fast.
- 7.4 When working on a separator with a hinged cover, ALWAYS be sure to lock the cover securely in the open position.
- 7.5 When using the disc stack compression tool as a lifting tool, ALWAYS follow the procedure exactly as described in the Maintenance & Repair manual.
- 7.6 Handle components with care. Do not place parts directly on the floor. Use a clean rubber mat, fibre-board or a wooden pallet. Protect all parts against damage, dust and dirt. Make sure that they are clean and free from burrs when assembling.
- 7.7 Do not use high pressure hoses or excessive amounts of water when cleaning the inside of the bowl casing as this can cause water to enter into the spindle bearings.

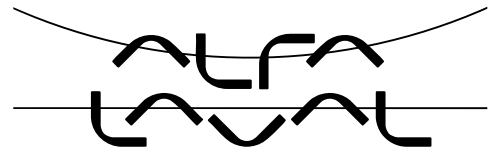
NEVER hose down a separator with a direct water jet. Totally enclosed motors can



be damaged by direct hosing to the same extent as open motors. Be careful even when the motor is equipped with a protecting hood. NEVER direct the water jet on the ventilation grills.

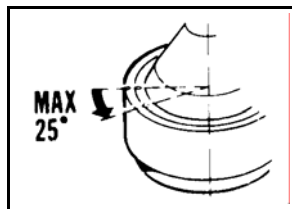
- 7.8 Do not clean bowl parts with solutions that may be corrosive to the particular materials of these parts.
- 7.9 When assembling a bowl, always make sure that the threads on bowl components are thoroughly clean, free from grit and properly lubricated.
- 7.10 A separator bowl is balanced as a complete unit. Components of one bowl must not be interchanged with those of another bowl (unless definitely permitted). Make sure that no parts are left out during assembly and that parts with alignment marks are positioned correctly. All major parts are marked with the full serial number or the last three digits for identification purposes.
- 7.11 NEVER start a separator without its bowl. This might damage the bearings.



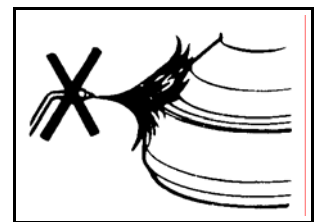


## 8. MAINTENANCE

- 8.1 ALWAYS use the proper equipment such as chain hoists, dollies, etc., when lifting and/or moving heavy components such as motors, bowl bodies, bowl assemblies, etc..
- 8.2 At each service, or at least every third month, the most important parts should be checked for fatigue damage. Special attention should be given to the bowl pillars for the solids discharge ports, nozzles, threads of bowl body/large lock-ring as well as the frame and the upper frame part which are hit by ejected solids and/or operating liquid. If process or cleaning liquids are corrosive or erosive the inspection interval must be shortened.
- 8.3 Note that the assembly mark  $\Phi$  on the main lock ring must be aligned or pass the  $\Phi$  mark on bowl body or bowl hood when the lock ring is fully tightened. In this position there must be proper compression of the disc stack. Check that all parts are securely tightened. If the  $\Phi$  mark on the large lock ring passes the corresponding mark on the bowl body/bowl hood by more than 25 degrees, do not

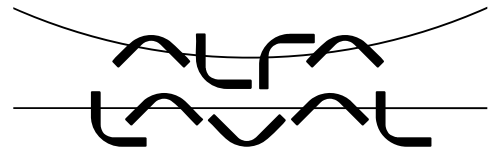


- operate the separator but consult your ALFA LAVAL representative.
- 8.4 Ensure that the bowl is fitted with the correct number of discs to obtain appropriate disc stack compression. If you remove a disc ALWAYS replace it with a new one to avoid unbalance.
- 8.5 NEVER heat bowl parts, such as bowl body, bowl hood, lock rings, etc., with a naked flame or attempt repairs by welding. This could destroy the mechanical and structural strength of the material.
- 8.6 On separators equipped with a friction brake, make sure that the pad is in a good condition.
- 8.7 Many separators are equipped with a paring disc liquid discharge. It is important that the paring device/feed tube assembly has the correct height adjustment. It is especially important to check this on new or overhauled separators.
- 8.8 Periodically inspect and test by operating all the automatic shut-off devices and monitoring systems provided to make sure they will function correctly when required.



**IF YOU ARE IN ANY WAY UNCERTAIN, CONSULT YOUR ALFA LAVAL REPRESENTATIVE!**

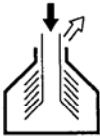
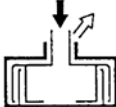
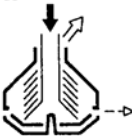
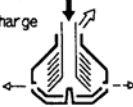
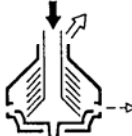
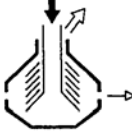
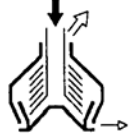
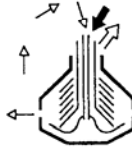
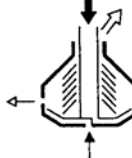
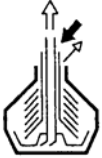
For reliability and maximum safety we strongly recommend that an ALFA LAVAL Service Engineer inspects your separator at regular intervals, recommended by the ALFA LAVAL Service Engineer.



# APPENDIX A: BOWL TYPES

The following figure shows schematic characteristics of different bowl types. The bowl type is stated in the separator designation, in most cases as characters no. 3 and 4.

**Example:** CHQX 320 B-31CGR.

|                   |                                                                                                                                         |                                                                                                                                                                                       |                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Solid Wall Bowl   | B Base design<br>                                      |                                                                                                                                                                                       | K Cylinder Inserts<br>                            | <u>Table of signs</u><br> Main Inlet<br> Main outlet<br> Continuous flow solids<br> Intermittent flow solids<br> Auxiliary liquid |
|                   | PX Base design. Radial peripheral ports and slide<br> | KX Peripheral ports and slide with nozzles for both continual and intermittent solids discharge<br> | AX Axial peripheral ports and slide<br>          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Bowl with Nozzles | SX Base design. Peripheral nozzles<br>               | DX Nozzles at reduced diameter<br>                                                                 | QX Peripheral nozzles. Solids recirculation<br> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|                   | TX Peripheral nozzles. Wash water feed<br>           | UX Internal nozzles with paring device<br>                                                         |                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

## **2. Compliance Declaration**

### **2.1 EU declaration**

### **3. General Information**

**3.1. Quality specification of water - 553406 R10**

**3.2. Quality specification of air - 553407 R6**



|                             |               |                 |
|-----------------------------|---------------|-----------------|
| Document Title              | Document ID   | Version         |
| Quality specification water | 553406        | 10              |
| Issued by                   | Department    | Page            |
| Andreas Rundby              | HSS           | 1(1)            |
| Approved by                 | Approval date | Approval Status |
| Anna-Karin Karlsson         | 2013-04-16    | APPROVED        |

Operating water is used in the separator for several different functions: e.g. to operate the discharge mechanism, to lubricate and cool mechanical seals.

Poor quality of the operating water may cause erosion, corrosion and/or operating problem in the separator and must therefore be treated to meet certain demands.



**Alfa Laval accepts no liability for consequences arising from unsatisfactory purified operating water supplied by the customer.**

The following conditions must be fulfilled:

1. Turbidity free water, solids content <0.001% by volume.  
Deposits must not be allowed to form in certain areas in the separator.
2. A maximum particle size of 50 µm.
3. A total hardness of less than 180 mg CaCO<sub>3</sub> per litre, which corresponds to 10°dH or 12.5°E.  
Hard water may with time form deposits in the operating mechanism. The precipitation rate is accelerated with increased operating temperature and low discharge frequency. These effects become more severe as the hardness of the water increase.
4. A chloride content of maximum 100 ppm NaCl (equivalent to 60 mg Cl/l).  
A chloride concentration above 60 mg/l is not recommended.  
Chloride ions contribute to corrosion on the separator surface in contact with the operating water, including the spindle. Corrosion is a process that is accelerated by increased separating temperature, low pH, and high chloride ion concentration.
5. pH>6  
Increased acidity (lower pH) increases the risk for corrosion; this is accelerated by increased temperature and high chloride ion content.



|                                                    |  |                                    |                                    |
|----------------------------------------------------|--|------------------------------------|------------------------------------|
| Document Title<br><b>Quality specification air</b> |  | Document ID<br><b>553407</b>       | Version<br><b>6</b>                |
| Issued by<br><b>Andreas Rundby</b>                 |  | Department<br><b>HSS</b>           | Page<br><b>1(1)</b>                |
| Approved by<br><b>Anna-Karin Karlsson</b>          |  | Approval date<br><b>2013-04-16</b> | Approval Status<br><b>APPROVED</b> |

The supply of compressed air to separator discharge system, valve actuators, positioners, instruments etc. must be of such a quality that satisfactory function is ensured for a reasonable time.



**Alfa Laval accepts no liability for consequences arising from unsatisfactory purified compressed air supplied by the customer.**

The following conditions must be fulfilled:

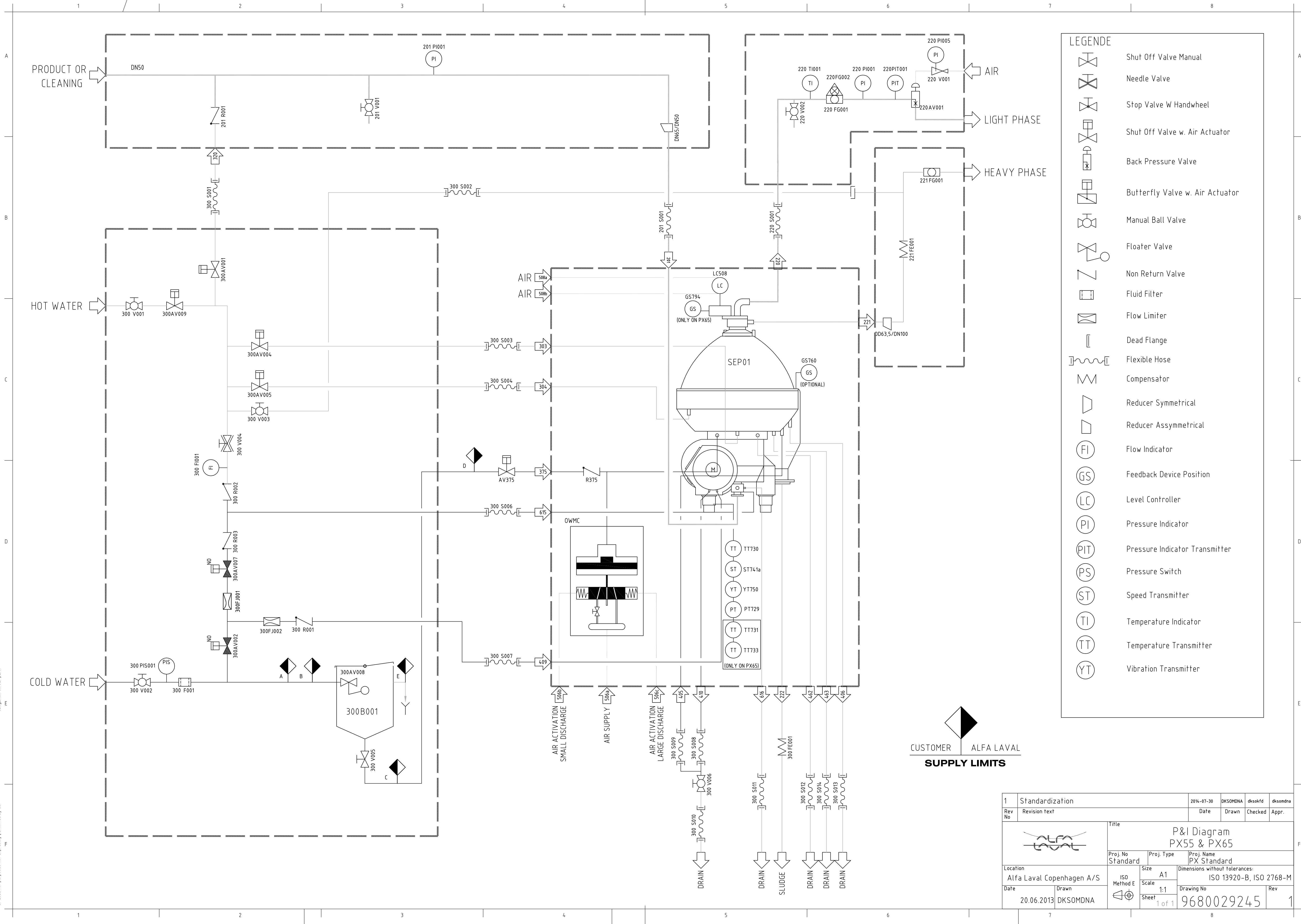
1. Dirt in the form of solid particles down to a size below 10 micron (0.01 mm) must be removed from the air.  
Removal of dirt from the air is preferably done by means of special filters or reducing valves provided with filters.  
Air filters should be easily surveyable and accessible in order to facilitate daily condition checks, and exchange of the filter cartridge.
2. Oil must be removed from the air.  
Oil is always transferred to the compressor air from oil-lubricated compressors and must be removed to the highest possible degree. Oil constitutes a serious contamination, which is difficult to remove from the instruments.  
Removal of oil from the air should be done by means of special filters or oil separators before the instruments. In small plants oil-free compressors can be used as an alternative.
3. At the inlet to an instrument, the dew point of the compressed air should lie at least 10 °C below the lowest ambient temperature.  
A correct dew point is usually obtained by using an absorption drier of suitable capacity. If the air humidity is high a primary separator should be used before the filter.  
In the compressed-air system a condensation takes place at various rates, depending on the moisture content at the air inlet, the temperature before and after the compressor and partially lower temperature in any cold zone passed by the pipe. The air must be dried with regard to the lowest temperature existing after the drying device, so that condensate in the instruments is avoided. Note that the air will also be cooled through expansion after passing constrictions and nozzles in the instruments, with condensation as a result.

## **4. Process Information & Drawing**

### **4.1. P&ID - 9680029245 R01**

Only to the extent expressly agreed in writing, this document may constitute a contractual obligation on our part.

This document and its contents are our exclusive property and no part thereof may be transmitted or communicated to any other party or used for purposes not expressly permitted by us.



---

## **5. Installation Manuals**

### **5.1. Instruction book - 9034330 02 R01**